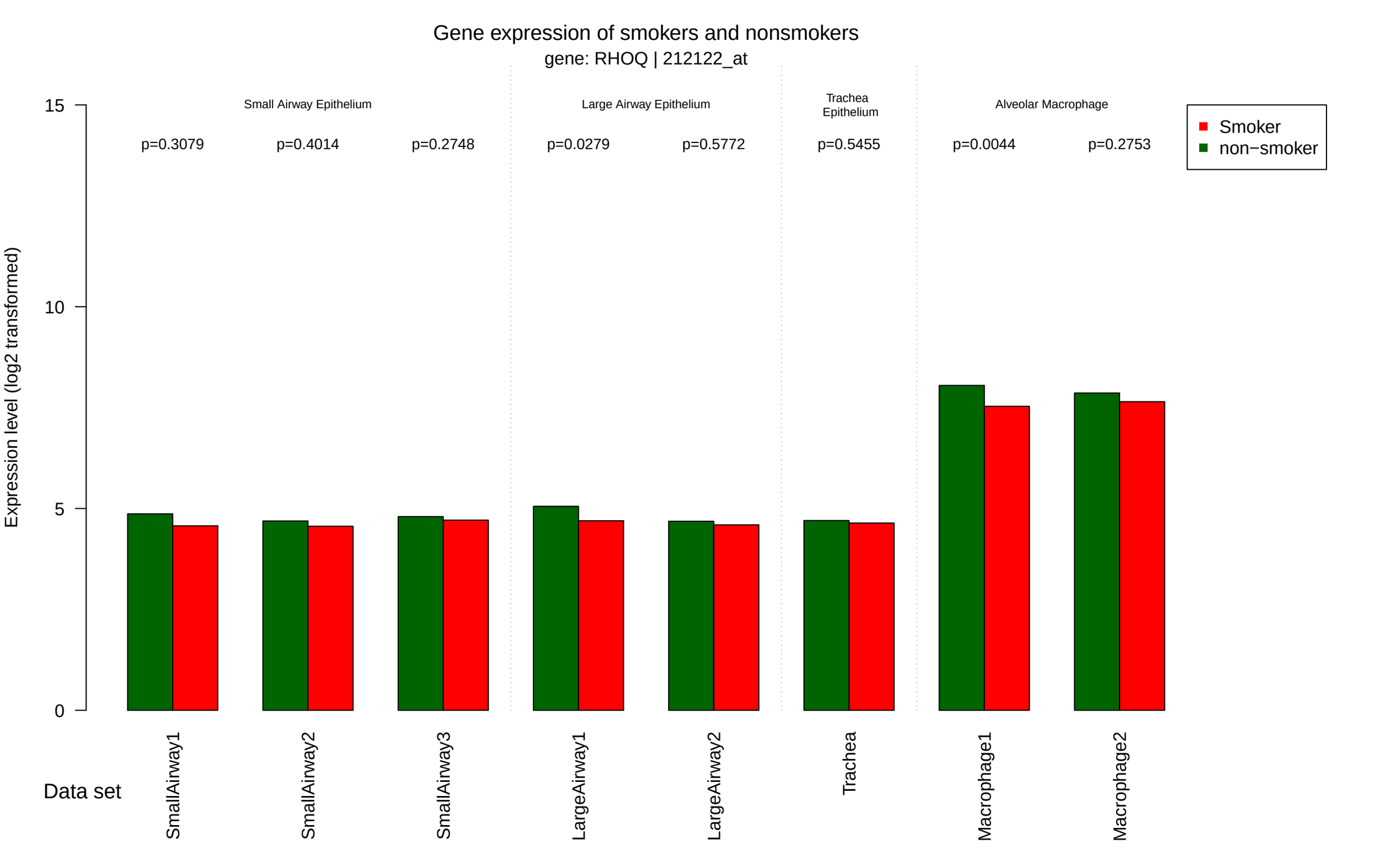
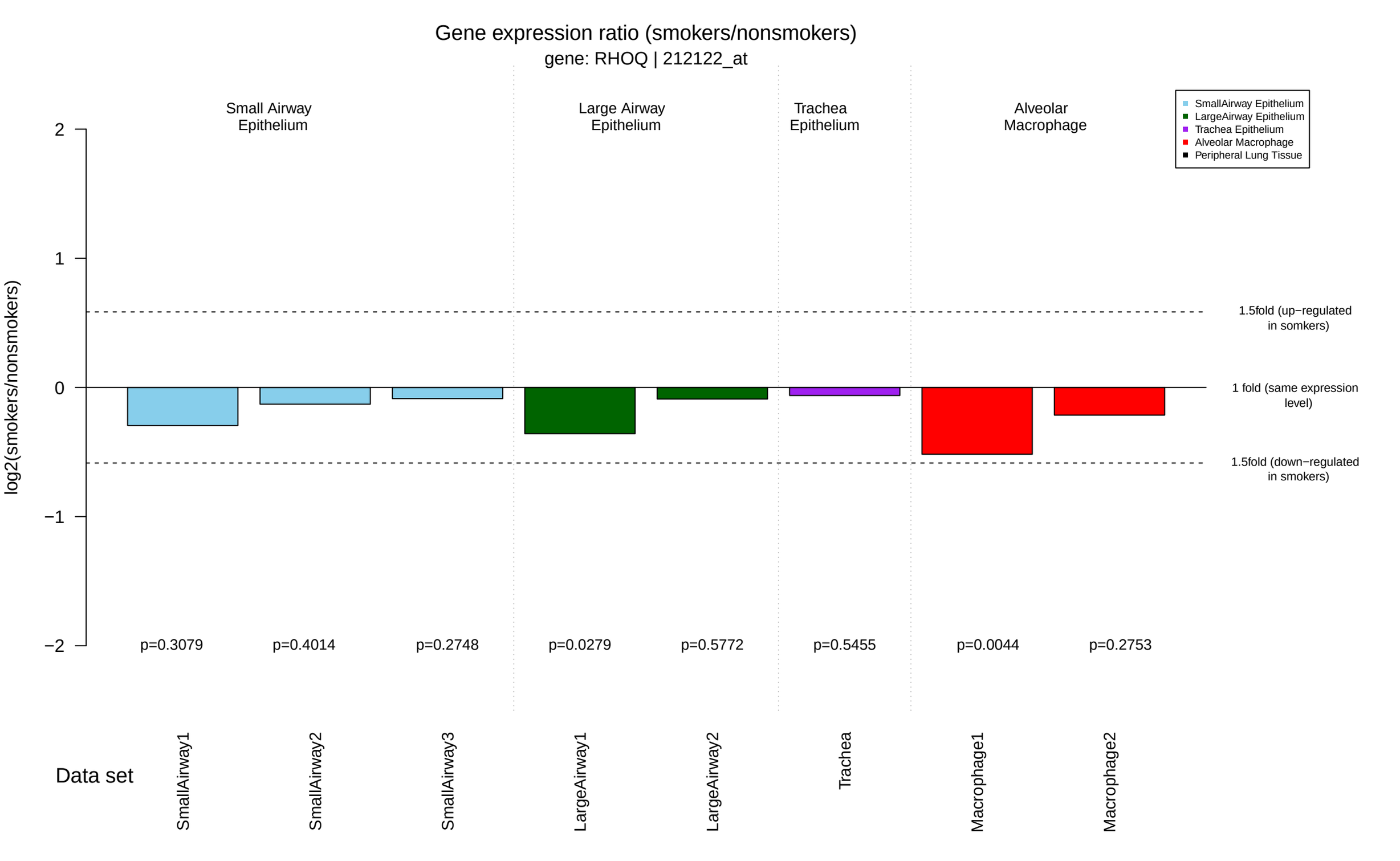
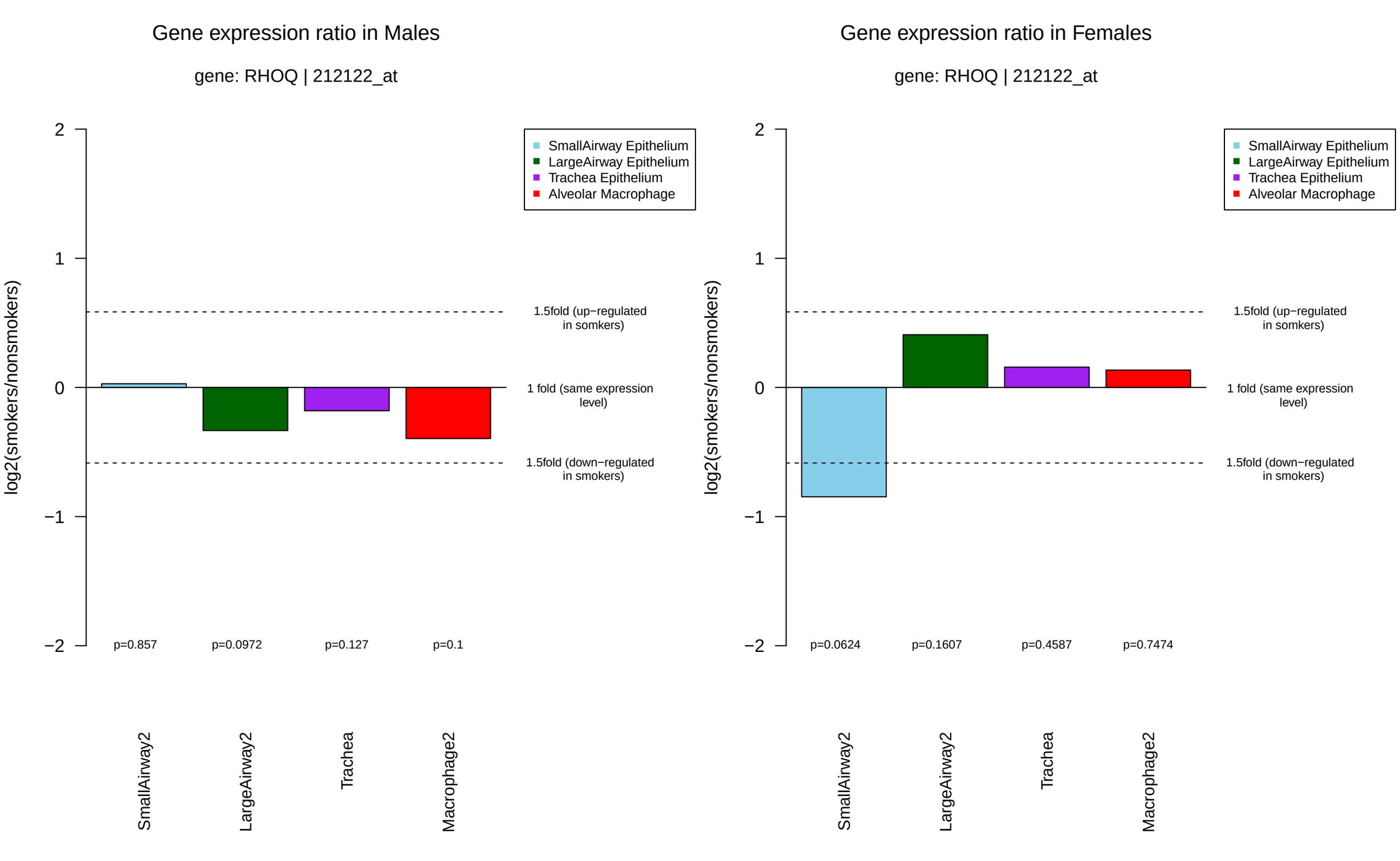
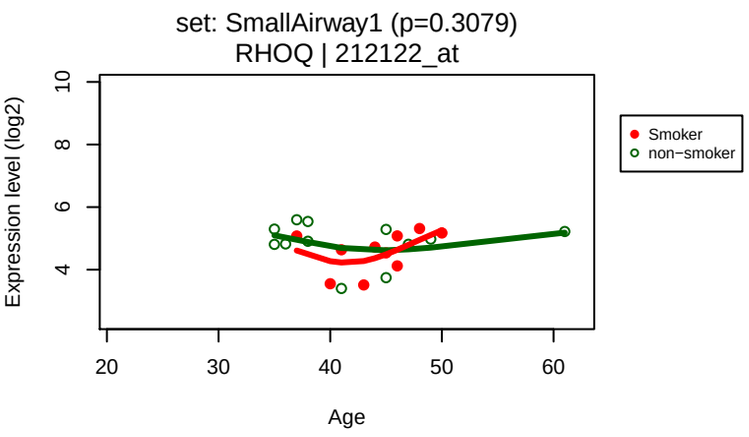
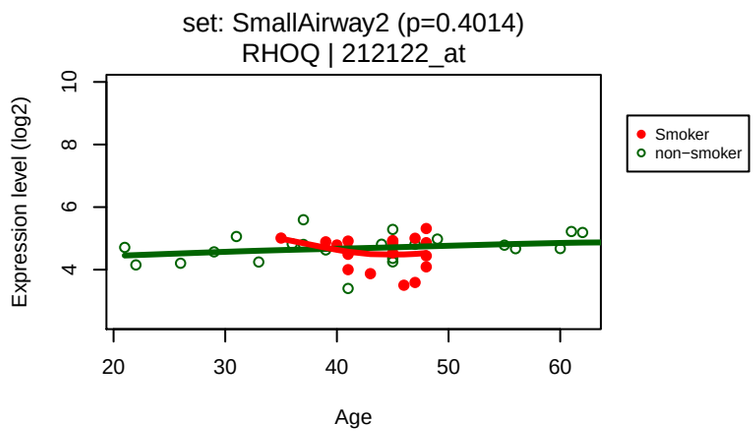
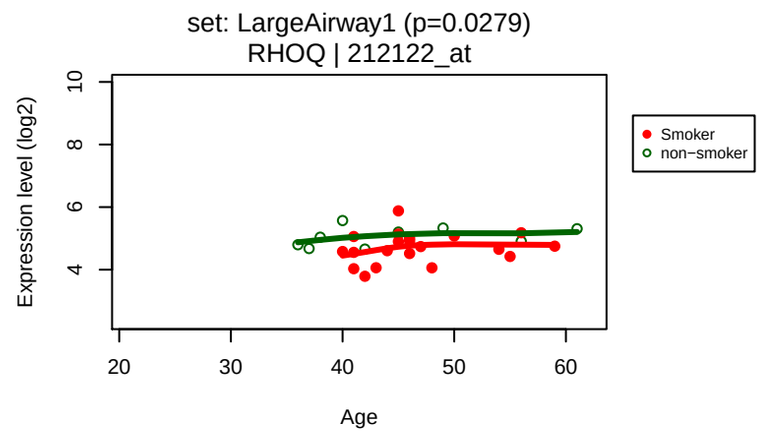

Gene expression of smokers and nonsmokers
gene: RHOQ | 212122_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3079
p=0.4014
p=0.2748
p=0.0279
p=0.5772
p=0.5455
p=0.0044
p=0.2753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

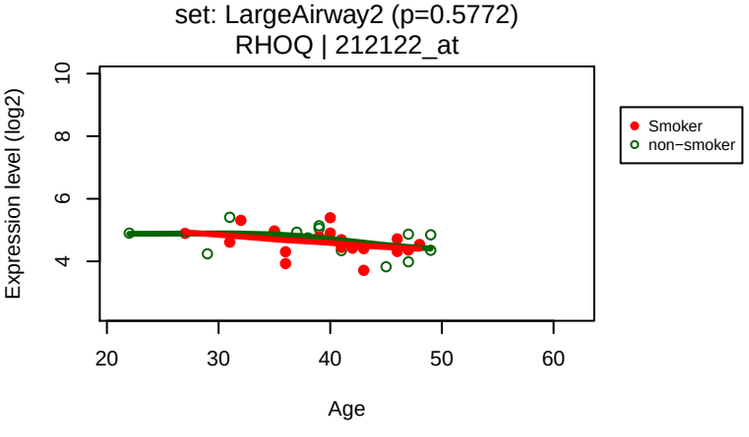

Gene expression ratio (smokers/nonsmokers)
gene: RHOQ | 212122_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3079
p=0.4014
p=0.2748
p=0.0279
p=0.5772
p=0.5455
p=0.0044
p=0.2753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

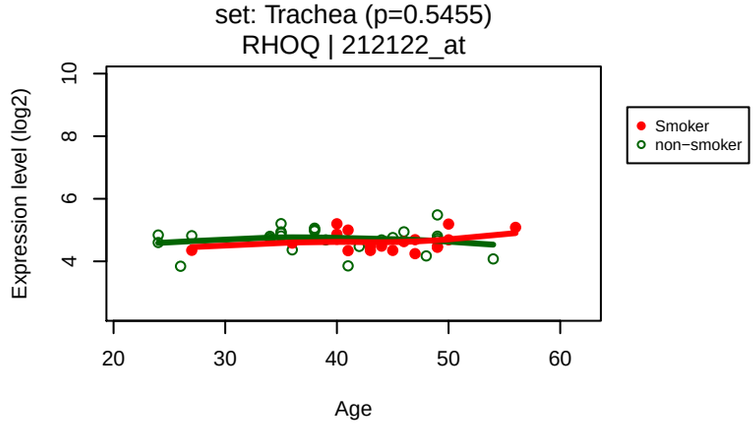

Gene expression ratio in Males
gene: RHOQ | 212122_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.857
p=0.0972
p=0.127
p=0.1
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOQ | 212122_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0624
p=0.1607
p=0.4587
p=0.7474
SmallAirway2
LargeAirway2
Trachea
Macrophage2

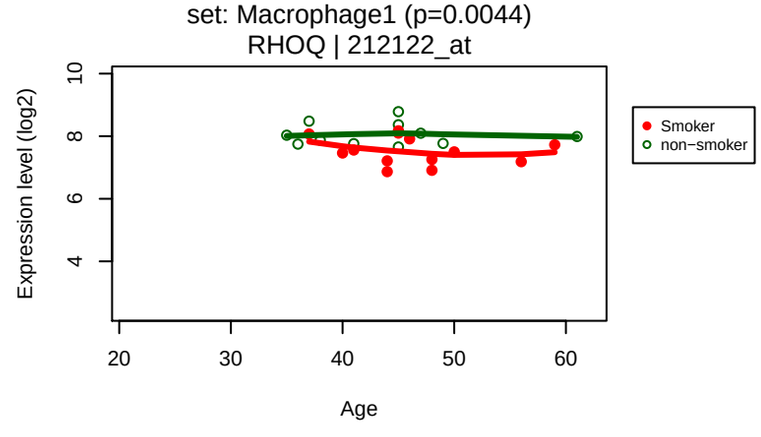

set: SmallAirway1 (p=0.3079)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker

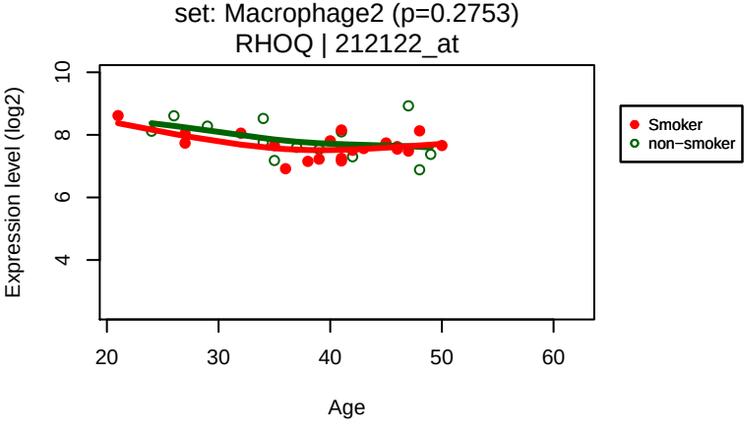

set: SmallAirway2 (p=0.4014)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker

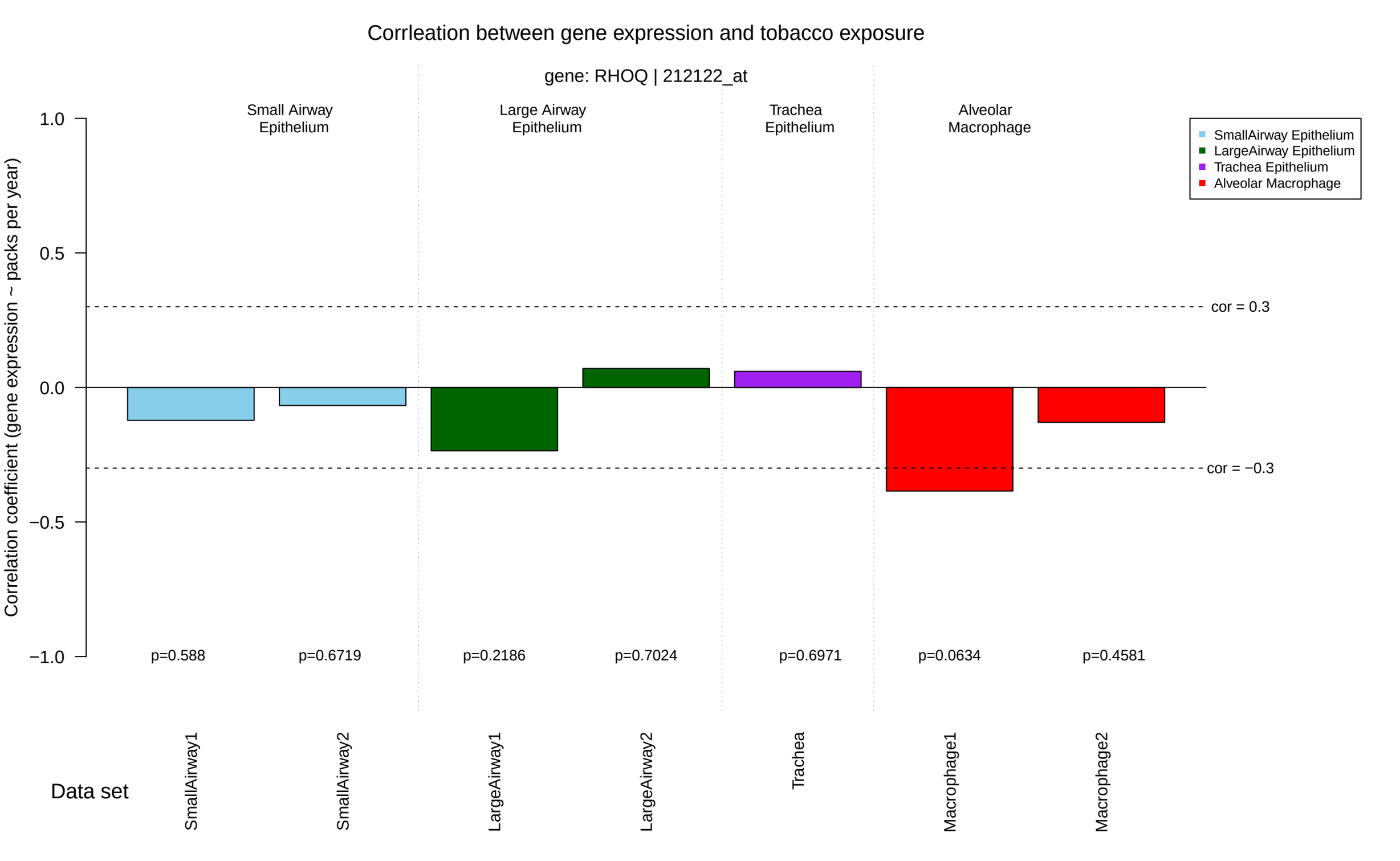

set: LargeAirway1 (p=0.0279)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5772)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5455)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0044)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2753)
RHOQ | 212122_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOQ | 212122_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.588
p=0.6719
p=0.2186
p=0.7024
p=0.6971
p=0.0634
p=0.4581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2